

Harun Adiyaman

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EDUCATION

Virginia Tech

M.S. in Computer Engineering

B.S. in Computer Engineering: Networking & Cybersecurity, Software Systems: 3.51/4 GPA

Blacksburg, Virginia

Expected 2027

Graduated May 2025

SKILLS

Programming: Python, C, C++, Rust, x86 Assembly, Bash, Object Oriented Programming, Multithreading

Reverse Engineering: Ghidra, Binary Ninja, Wireshark

Networking: Socket Programming, TCP, UDP, IP

Testing: Catch2, PyTest, Cargo, Unit & Integration Testing

Software: KiCad, VirtualBox, Vim, Tmux, VSCode, Eclipse

Tools: Git, Linux, Docker, C/Make, GDB, KiCad, QEMU, FreeRTOS, Agile

EXPERIENCE

MIT Lincoln Labs

Summer Research Intern – Open & Embedded Systems Group

Boston, Massachusetts

May 2025 – Present

- Developed a playback **simulator** in **C++** for an EO/IR sensor to facilitate system integration testing by removing hardware from the loop.
- Ported C++ radar signal processing kernels to **Rust** to evaluate performance with unit & integration **testing**.
 - Identified performance bottlenecks with profiling tools, achieving improvements.
- Presented project findings to technical teams, leads and stakeholders.

Virginia Tech Major Design Experience

High Altitude Ballooning at Virginia Tech - <https://github.com/eiharun/SP25-20>

Blacksburg, Virginia

Sep 2024 – May 2025

- Led a 5-member capstone team using agile workflow to design, prototype, and test a motorized venting system with a LoRa-based command-and-control link, enabling reliable balloon venting at altitudes exceeding 80,000 ft.
- Collaborated with the customer and subject matter experts to refine project requirements, acquire necessary skills, and ensure alignment with objectives, while regularly updating key stakeholders.

Virginia Tech Independent Study @ MICS

Wireless Gait Sensor Development

Blacksburg, Virginia

Sep 2024 – May 2025

- Engineered a low-power compact PCB using the TI CC2640R2F MCU and powered by a coin cell battery, to interface with pressure sensors and transmit gait data via Bluetooth in order to predict and prevent the falls of elderly people.

Virginia Cyber Range

Capture The Flag CTF Developer

Blacksburg, Virginia

Jan 2023 – May 2025

- Developed and deployed over 50 various technical challenges in reverse engineering, networking, and cybersecurity for CTF competitions at the high school and college level.
- Communicated complex technical challenges clearly to diverse audiences including competitors, instructors, and supervisors. Mentored incoming interns.
- Authored detailed documentation highlighting solutions and theory behind each challenge.

PROJECTS

PinguOS - <https://github.com/eiharun/PinguOS>

Aug 2025 – Present

- Developed a 32-bit x86 (i386) operating system from scratch, implementing a **bare-metal C++/assembly** kernel, Multiboot-compliant GRUB bootloader, and bootable ISO; tested and debugged using **GDB & QEMU**, gaining hands-on experience with kernel development, ELF binaries, bootloaders, and low-level system initialization.
- Configured GDT/IDT and built **interrupt**-driven keyboard/mouse device drivers with event-driven input handling, enabling low-level I/O, hardware abstraction, and real-time VGA text-mode rendering.
- Designed and integrated a **preemptive multitasking scheduler** to manage concurrent processes and context switching at the kernel level.
- Implemented dynamic memory management with a custom heap allocator, enabling allocation and deallocation of memory in a **bare-metal** environment.

Page Table Walker Linux Kernel Module

Apr 2025

- Designed a Linux kernel module that traverses page table entries for a specified virtual address, reporting physical addresses and flags at each level.
- Gained hands-on experience with low-level memory management in Linux, page table structures, and kernel module development.

Linux Scheduler Profiler Kernel Module

Mar 2025

- Developed a Linux kernel module to profile task scheduling events by tracking context switches and recording runtime statistics per process.
- Utilized kernel hooks and the /proc file system to expose profiling data to user space, enhancing understanding of scheduling behavior and kernel-level process management.

Modifying FreeRTOS Task Scheduling

Feb 2025

- Modified the FreeRTOS kernel to prioritize the execution of the lowest-priority task on startup, implemented periodic task scheduling logic on an Arduino Mega 2560, and analyzed and tested system behavior using serial communication.

Linux Kernel Custom System call

Feb 2025

- Modified the Linux kernel by adding debug messages to system calls, implementing a custom system call for string encryption, and analyzing system call overhead. Developed and tested kernel patches for Linux v6.12, wrote a user-space test program, and measured performance.

Dots and Boxes Game

Nov 2024 – Dec 2024

- Managed a team of 3 to design and implement a modular real-time online game, designing an observable game model in Python OOP that interfaced with a scalable WebSocket server, REST API for user authentication and game management, MongoDB, and Dockerized deployment.

Concurrent Heapsort Algorithm

Nov 2024 – Dec 2024

- Designed a multithreaded concurrent heapsort algorithm, utilizing 16 threads to parallelize container sorting, improving performance by up to 90% for list sizes of 1 million and above, and reducing overhead for large data sets, as demonstrated by benchmark results.

Post-Lisp Language Interpreter & Drawing Application

Oct 2024 – Nov 2024

- Implemented a Postfix Lisp interpreter in C++ featuring tokenization, AST construction, and evaluation via post-order traversal, with a focus on modular design, and integration, functional, coverage, and unit testing using Catch2.
- Developed a GUI drawing application using Qt that extends the post-lisp language interpreter to draw graphics.

NetClock Server

Oct 2024 – Nov 2024

- Developed a multi-threaded Python server managing synchronization requests from IP clocks using a UDP protocol, ensuring efficient handling of concurrent clients, non-blocking client registration persistence, and real-time updates, while integrating object-oriented design, logging, and Docker container deployment.

CheapTalk Messaging Server

Sep 2024 – Oct 2024

- Designed a TCP-based messaging application utilizing sockets in a team of 3 with a custom protocol supporting asynchronous communication, multi-channel messaging, and dynamic channel creation. Optimized scalability and performance using input multiplexing and a single-threaded architecture.

Virginia Tech AMP Lab

Apr 2024 – Aug 2024

- Constructed a simple GUI on an SPI OLED Display with directional buttons on an STM32 microcontroller, and interfaced with a 6-axis gyroscope module using embedded programming in C.

ACTIVITIES AND LEADERSHIP

Deloitte Cyber Threat Competition

Fort Worth, Texas

Technological Lead

Feb 2023

- Incident Response Wargame Simulation: Led 4-person team in incident response wargame, presenting to simulated executive leadership.

CCI Cyber Camp Summer

Arlington, Virginia

Member

Jun 2022

- XG Testbed: Worked in a team to build a small 4G LTE Cellular Network | AI Workshop: Worked to build an AI in Python(scikit learn) to determine the most “tech-savvy” US state | Data Analytics Workshop: Used JMP to analyze TCP data in order to identify variables that determine malicious traffic.

CyberVT at Virginia Tech

Blacksburg, Virginia

Member

Nov 2022 – May 2023